

Shayan Majidy

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I am a quantum computing theorist working towards reliable quantum computers for scientific discovery. My research connects quantum error correction, the joint design of quantum algorithms and architectures, and quantum dynamics and thermodynamics—from understanding how quantum information behaves in physical systems to protecting it and putting it to use.

Interests: Quantum error correction; quantum algorithms and architectures; quantum dynamics and thermodynamics.

Research appointments

2026	Senior Research Scientist, QuEra
2026	Associate, Harvard University Adviser: Mikhail D. Lukin
2024–2026	Postdoctoral Fellow, Harvard University Advisers: Mikhail D. Lukin Michael Gullans Funding: <u>Banting Fellowship</u> , 2024–2026

Education

2019–2024	Ph.D. in Physics, University of Waterloo Advisers: Raymond Laflamme Nicole Yunger Halpern Funding: <u>Vanier Scholarship</u> , 2021–2024
2018–2019	M.Sc. in Physics, University of Waterloo Advisers: Raymond Laflamme
2011–2015	B.Sc. in Theoretical Physics, University of Guelph

Selected awards

Competitive fellowships

2024–2026	<u>Banting Postdoctoral Fellowship.</u> Government of Canada; CAD 140,000 over two years. Field-specific rank: 3rd of 185 applicants.
2021–2024	<u>Vanier Canada Graduate Scholarship.</u> Government of Canada; CAD 150,000 over three years.

Research honours

2025	<u>W.B. Pearson Medal.</u> University of Waterloo; creative doctoral research; normally one award per Science department annually.
2024	<u>John Brodie Memorial Award</u> (co-recipient). Perimeter Institute; recognizes creativity in doctoral research.
2022	<u>IQC Achievement Award.</u> Institute for Quantum Computing; excellence in graduate research.

Publications and preprints

* Indicates co-first authors.

1. **Achieving the limits of automorphism gates**
J. M. Koh*, **S. Majidy***, A. Chakraborty, A. Gong, S. J. S. Tan, N. Y. Yao
Preprint · [arXiv:2609.19250](#) (2026)
2. **Entangling logical qubits without physical operations**
J. M. Koh, A. Gong, A. C. Diaconu, D. B. Tan, A. A. Geim, M. J. Gullans, N. Y. Yao, M. D. Lukin, and **S. Majidy**
Preprint · [arXiv:2601.20927](#) (2026) · Selected for talks at [TQC 2026](#), [QEC 2026](#), [CQIQC-XI](#), and QPL 2026
3. **A fault-tolerant neutral-atom architecture for universal quantum computation**
D. Bluvstein, A. A. Geim, S. H. Li, S. J. Evered, J. P. Bonilla Ataides, G. Baranes, A. Gu, T. Manovitz, M. Xu, M. Kalinowski, **S. Majidy**, C. Kokail, N. Maskara, E. C. Trapp, L. M. Stewart, S. Hollerith, H. Zhou, M. J. Gullans, S. F. Yelin, M. Greiner, V. Vuletić, M. Cain, and M. D. Lukin
Nature **649**, 39–46 (2026) · [Journal](#) · [arXiv](#)
4. **Scalable and fault-tolerant preparation of encoded k -uniform states**
S. Majidy, D. Hangleiter, and M. J. Gullans
Physical Review A **112**, 042409 (2025) · [Journal](#) · [arXiv](#)
5. **Thermodynamic limitations on fault-tolerant quantum computing**
M. Bilokur, S. Gopalakrishnan, and **S. Majidy**
Preprint · [arXiv:2411.12805](#) (2024)
6. **Noncommuting charges can remove non-stationary quantum many-body dynamics**
S. Majidy
Nature Communications **15**, 8246 (2024) · [Journal](#) · [arXiv](#) · Selected for a talk at [CQIQC-X](#)
7. **Addressing misconceptions in university physics: A review and experiences from quantum physics educators**
S. Majidy
Preprint · [arXiv:2405.20923](#) (2024); revised 2025
8. **Noncommuting conserved charges in quantum thermodynamics and beyond**
S. Majidy, W. F. Braasch Jr., A. Lasek, T. Upadhyaya, A. Kalev, and N. Yunger Halpern
Nature Reviews Physics **5**, 689–698 (2023) · [Journal](#) · [arXiv](#)
9. **Critical phase and spin sharpening in SU(2)-symmetric monitored quantum circuits**
S. Majidy, U. Agrawal, S. Gopalakrishnan, A. C. Potter, R. Vasseur, and N. Yunger Halpern
Physical Review B **108**, 054307 (2023) · [Journal](#) · [arXiv](#)
10. **A unification of the coding theory and OAQEC perspective on hybrid codes**
S. Majidy
International Journal of Theoretical Physics **62**, 177 (2023) · [Journal](#) · [arXiv](#)
11. **Non-Abelian symmetry can increase entanglement entropy**
S. Majidy, A. Lasek, D. A. Huse, and N. Yunger Halpern
Physical Review B **107**, 045102 (2023) · [Journal](#) · [arXiv](#)
12. **How to build Hamiltonians that transport noncommuting charges in quantum thermodynamics**
N. Yunger Halpern and **S. Majidy**
npj Quantum Information **8**, 10 (2022) · [Journal](#) · [arXiv](#)
13. **Detecting violations of macrorealism when the original Leggett-Garg inequalities are satisfied**
S. Majidy, J. J. Halliwell, and R. Laflamme
Physical Review A **103**, 062212 (2021) · [Journal](#) · [arXiv](#)

14. **Exploration of an augmented set of Leggett-Garg inequalities using a noninvasive continuous-in-time velocity measurement**
S. Majidy, H. Katiyar, G. Anikeeva, J. Halliwell, and R. Laflamme
Physical Review A **100**, 042325 (2019) · [Journal](#) · [arXiv](#)
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Textbook

Building Quantum Computers: A Practical Introduction

S. Majidy, C. Wilson, and R. Laflamme
Cambridge University Press (2024).

An introduction to quantum computing hardware for advanced undergraduates and beginning graduate students. It compares physical implementations through shared concepts and notation and is used in university teaching internationally.

Research mentoring

Graduate research mentoring. Harvard University; University of Maryland; ETH Zürich; University of Waterloo.
Undergraduate research mentoring. Princeton University; University of Waterloo.

Research talks

Invited conference talks (1)

Achieving the limits of automorphism gates

22 Oct 2026 [PQTC 2026](#), Princeton

Contributed conference talks (7)

Entangling logical qubits without physical operations

2 Sep 2026 [TQC 2026](#), University of Sherbrooke

20 Aug 2026 [CQIQC-XI](#), University of Toronto

11 Jun 2026 [QEC 2026](#), Santa Barbara

Architectural mechanisms of a universal fault-tolerant quantum computer

8 Oct 2025 [Year of Quantum Across Canada](#), Perimeter Institute · [\[Recording\]](#)

Noncommuting charges can increase entanglement and induce critical dynamics

26 Aug 2024 [CQIQC-X](#), University of Toronto · [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

8 Dec 2022 [Quantum Non-Markovianity 2022](#), Rydges Newcastle (online)

Noncommuting charges: Bridging theory to experiment

22 Jul 2022 [Information Engines at the Frontiers of Nanoscale Thermodynamics](#), Telluride

Colloquia (5)

Fault-tolerant quantum computing: From fundamental limits to scalable architectures

30 Apr 2026 Colloquium, École de Technologie Supérieure

29 Apr 2026 Colloquium, University of Waterloo

16 Apr 2026 Colloquium, University of Alberta

20 Mar 2026 Colloquium, University of British Columbia

15 Jan 2026 Colloquium, Université de Sherbrooke

Invited seminars (23)

Entangling logical qubits without physical operations

- 26 Jun 2026 [Foxconn Seminar](#), Hon Hai Research Institute (online)
19 May 2026 [VTQ Seminar](#), Virginia Tech
13 Apr 2026 [QASAR Seminar](#), Inria (online)
4 Mar 2026 [QuICS Seminar](#), University of Maryland · [\[Recording\]](#)
7 Jan 2026 Quantum Information Seminar, Perimeter Institute · [\[Recording\]](#)

Architectural mechanisms of a universal fault-tolerant quantum computer

- 3 Oct 2025 [CQIQC Seminar](#), University of Toronto · [\[Recording\]](#)

A noncommuting charge puzzle

- 8 Jan 2025 Quantum Information Seminar, Perimeter Institute · [\[Recording\]](#)
28 Jun 2024 Harvard Quantum Initiative Seminar, Harvard University

Noncommuting charges can increase entanglement and induce critical dynamics

- 16 Jan 2024 [Yale Quantum Institute Seminar](#), Yale University
29 Nov 2023 Quantum Information Seminar, Perimeter Institute · [\[Recording\]](#)
22 Sep 2023 Theory Seminar, Princeton University

Monitored quantum circuits with noncommuting conserved quantities

- 15 Sep 2023 Qiskit Seminar, IBM (online) · [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

- 22 Jun 2023 [RQS Annual Workshop](#), University of Maryland
28 Mar 2023 InfoQ Seminar, Université de Sherbrooke
24 Mar 2023 Special INTRIQ/CPM Seminar, McGill University
24 Feb 2023 Institute for Theoretical Physics Seminar, Stanford University
22 Feb 2023 Pitzer Center Seminar, University of California, Berkeley
22 Feb 2023 [Redwood Seminar](#), University of California, Berkeley
7 Feb 2023 Quantum Field Theory and Gravity Seminar, Universität Würzburg (online)
3 Feb 2023 [CQIQC Seminar](#), University of Toronto

Noncommuting charges: Bridging theory to experiment

- 18 Aug 2022 Theoretical Physics Seminar, Australian Institute of Physics (online)
2 Aug 2022 RQS Seminar, University of Maryland

Noncommuting charges: Bridging theory to experiment

- 9 Jun 2021 Quantum Information Theory Seminar, University of Bristol (online)

Teaching and educational development

Independent course instruction

Fall 2022 **Sessional Lecturer**, University of Waterloo.
Introduction to Implementation of Quantum Information Processing, fourth-year course.
Independently taught the course; redesigned the syllabus, teaching methods, and assessments; delivered lectures and coordinated teaching assistants.
Course evaluation: 4.8/5 · 97% response rate.

Teaching assistantships

2020–2022 **Teaching Assistant**, University of Waterloo.
Quantum Information Processing Devices (Winter 2020, 2021, 2022); *Introduction to Implementation of Quantum Information Processing* (Fall 2021); *Electricity and Magnetism 1* (Winter 2021).
Led discussion sessions, graded assignments, and held office hours.

Educational development

Jan–May 2022 **Graduate Educational Developer**, Centre for Teaching Excellence.
Jan–Dec 2021 **TA Workshop Facilitator**, Centre for Teaching Excellence.
Across both roles, designed and facilitated workshops, conducted teaching observations and microteaching sessions, and provided individual feedback to graduate instructors.

Teaching credentials

Certificate in University Teaching, University of Waterloo.
Three-course program (typically 6–9 terms): teaching workshops, reflective writing, a pedagogical research project, teaching dossier, and two observed teaching sessions.

Professional service and outreach

Peer review

Journal reviewing (43 reviews). *PRX Quantum* (8); *Physical Review Letters* (4); *Nature Communications* (1); *npj Quantum Information* (3); *Physical Review Research* (2); *Physical Review A* (24); *Physical Review E* (1).
Conference subreviewing. QIP 2026; QEC 2025; QCTIP 2025; TQC 2024.

Scientific event organization

2023 **Organizer**, Raymond Laflamme’s 60th Birthday Conference.
University of Waterloo.
2021 **Seminar organizer**, Quantum Steampunk Seminars. University of Maryland.
2020 **Organizing committee**, Canadian Graduate Quantum Conference. University of Waterloo.

Selected outreach and public engagement

Founder, Unentangled. Built a nonprofit bringing current research into high-school classrooms. At its peak, the program served more than 1,000 students annually with a team of 10 volunteers. [\[Documentary\]](#)
Public engagement: Invited panelist, *Quantum + The Near Future*, Quantum Connections (2024); interview, CBC Radio’s *Quirks & Quarks* (30 Dec 2023).
Research coverage: Phantom codes featured in [New Scientist](#).